Placement Diary

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Evidence of Student Experience

This diary presents activities for each week of the placement, and includes comments on the experience gained by undertaking each activity.

Diary

Week 1 – 24/07/2023

- Obtained the code-base.
- Examined the Jomini Engine codebase in an attempt to piece together the organisation and functionality of the project.
- My experience in systems running servers written within a C# environment is minimal and as such this project has exercised my abilities to learn this new system

Week 2 – 31/07/2023

- Delved deeper into the Hundred Year War gaming system while attempting to piece together which functionalities exist within the Jomini Engine currently.
- The current implementation of the client is not behaving as expected within my development environment. I have made some progress in fixing this but the system is still experiencing problems, the server side implementation appears to be running fine however.
- I am anxious about judging the current state of the project as the progress currently can only be gauged from reading the scripts associated as the current development environment is not compatible with the client as of writing. The process has however allowed insights into the Unity engine and its behaviours towards external packages.

Week 3 – 07/08/2023

- Research into the role of religion with the main focus of discovering opportunities to improve the existing systems or implement new features.
- Explored the Pope player's abilities to progress NPCs through several roles relating to their status within the clergy as well as admittance to the clergy itself.
- Focus on two specific player characters which begin as Archbishops within the initial game state however they are not currently implemented within the game, looking to focus on preparing the system to implement these player characters and what is needed to support the player character actions.
    - The actions decided upon relate to their ability to set tax rates within fiefs they own as well as their interactions with the special tax benefits that the clergy grants.
        - Not entirely clear on the tax benefits as of the current moment however will update in following diary entries with the progress.
- Encountered further problems with running the Jomini Client, I believe that the problem may be with the authentication process not allowing the correct information to be served to the client to progress past the login screen. Additional theory is that loading the map data itself is causing the client to hang.

Week 4 – 14/08/2023

- Monday

        - Continued work on the client in search of a fix to the client loading issues.
        - Currently still experiencing freezing behaviour when trying to log into an account.
        - Note that running the map scene itself will not cause any freezing behaviour so current belief is that communication between the server and client is causing the issue.
- Tuesday
    - I have been struggling with comprehending the large amount of information in each part of both the client and server files so have begun to write up a large scale documentation of the system so that I can understand how each piece interacts and communicates with each other.
    - The goal with this is that I can work more efficiently as well as having something that can be passed onto future developers.
- Wednesday/Thursday
    - Working on the client login and noticed that a few issues may be being caused by the log in functions.
    - Currently the system is rejecting all login with the reason of invalid username. The way that the functions are written lead me to believe there should be an auto login procedure that is not functioning correctly for one reason or another.
    - The issue could be to do with the package that is being used for the log in functionalities - Ledgrin.

Week 4 – 21/08/2023

- Solved the problem with the login system
- Attempted a fix on the standalone version, compilation errors.

Week 5 – 28/08/2023

- Began a documented effort at cataloguing different bugs in the current build.
- Relating to one of the bugs that was discovered is that the code within the project seems to refer to an ‘auto login’ functionality which would allow a user to play without entering a username or password. Currently an attempt to do this will result in the client trying to force a change to the map scene, in order to initialise the scene it requires information from the server however the server in its current state will not allow an anonymous user to view any of the data.
    - Progress for a fix would start on the server side
- Some fief interactions seem to be missing or broken, in future look into
    - Fief Pillage
    - Fief Siege
    - Fief ownership transfer
- Other minor bugs are visual in nature such as text boxes not sizing correctly and overlapping textures.

Week 6 – 04/08/2023

- Working on the missing fief interactions, scope is currently on the overlord actions that can be taken within a fief.
- The first task which is in progress is the ability for an overlord to transfer ownership of a fief to an eligible noble.
    - In the progress of this task it’s been noted that a large chunk of the code has been deprecated within unity so while all tasks are being completed a mild clean up is underway.
        - Additional note: Since a rewrite was in progress prior to acquiring the project this is not a high priority goal.
- Compilation Errors currently, the deprecation within ViewFief scripts causing issues.

Week 7 – 11/08/2023

- Compilation errors have been dealt with, there were some issues with the package manager in unity when the project had been initially pulled

Week 8 – 04/09/2023

- Illness complicated the progress of the project so the contributions were lighter on this week.

- Used the time available to review the information that would be relevant for implementing a religion and faith system.

Week 9 - 11/09/2023

- Began the work on the overlord menu that would allow for a user to manage their owned fiefs and perform various actions.

Week 10 - 18/09/2023

- Continued to work on the Overlord menu
    - Some trouble in the last few days of the week with the ProtoMessageClient. I have not completed this component but understand the flow of data between the client and server better.
        - Need to add more documentation to the project to further clarify for future development.
- Noted some changes that could be made to the file structure of the project.
    - Removal of redundant files
    - Renaming/Restructuring for greater clarity
    - Assumption that this would be handled in the rewrite.

Week 11 - 25/09/2023

- Fixed the issues with the github repository and created a remote repo which can be used for collaborative work.
- Updated the README.md to better guide on the installation of the project.
- Began work on the faith system.
    - Introduced new elements to the game objects collection
        -

Religion	Faith_A	Christian
Religion	Faith_B	Celtic
Sect	Faith_A1	Roman
Sect	Faith_A2	Poly
Sect	Faith_B1	Scottish
Sect	Faith_B2	Irish

            - Not accurate to reality with the described faiths, mainly for the testing of functionality
        -

0 pc	Char_300	Joann	Stratford

            - A playable archbishop character with an associated player account
        -

```
// test account for the religion system
LogInManager.StoreNewUser("faithTest", "faithtest");
```

    - Further information will be in the Religion and Faith document.

Week 12 - 02/10/2023

- Reconsidered direction with the religion and faith system.
- Attention pointed towards the heresy subsystem which would allow for interaction between the clergy, church and the rest of the game world.
- The religions present have also been simplified in concept. At the moment there is one main/base religion from which different doctrines can be derived from. An example of this is the baseline of Roman Catholicism and the Three Nails interpretation.
- The current system of the csv import has required some changes in the server side which is currently being looked at.

Week 13 - 09/10/2023

- Monday
    - Came across some trouble in the csv import file, began an effort at documenting how the data flows through the program to help focus the work into a more effective process.
    - Wrote up the information about the organisations.

        - Still need to do a full in depth description of the changes needed to be made in the system side of things for the organisation but some notes have been made.
- Tuesday/Wednesday
    - Joint effort between the two days to overhaul the overlordship menu into one that contains both the individual overlordship but also the organisations that an individual could be a part of.
- Thursday
    - Switch character menu is better for switching into a general character menu where they can access fiefs and organisations

Week 14 - 16/10/2023

- Figured out that the error with the csv files i've been encountering is due to microsoft excel changing the formatting when editing the file. Note that if changes are to be made to any of the initial csv files they must be made in notepad to avoid extra delimiters being added.
- Implemented a test action button that will help show the process of how actions can be added to the game. This is currently being used to test the faction request functionalities.
- Factions are beginning to be modelled, currently they exist as both factions and faction requests.
    - The idea behind this is that faction requests can be created by an overlord which would appear publicly within a list that could be viewed by all players in the game. Players can then either join a faction or sign onto a faction request to be a founding member.
- Having trouble with the new protomessage inherited type that handles the protomessage for the faction requests.

Week 16 - 30/10/2023

- This week I had some trouble with setting up the faction and organisation systems. The creation of new proto objects was still causing confusion, in previous work I had been working with functions and objects created prior to my involvement.
- I took some time to step away from the project and research into protocol buffers as I believe that my problems were stemming from a fundamental misunderstanding of how to utilise protocol buffers to serialise and stream information between the client and server.
- The creation of new proto objects was causing additional problems due to unfamiliarity with how Unity utilises dynamic link libraries, this is the first project where I've had to deal with external tools for unity development that could not be accessed through the Unity package manager. I have now figured out this side of the Unity development environment.
- Main Learning Points:
    - Protocol Buffers
    - Dynamic Link Library Compilation within Unity development environment
- Areas To Look Into Further:
    - Riak Distributed Systems - Goal before I depart the project is that I want to have this project tested on distributed systems and so next week I plan to look into how I would begin this work.
    - Communication Skills - This is a more abstract thing I need to work on, I found that when I was having a solo meeting with Yuna I had some trouble communicating some ideas I had. I need to get more practice with communicating technical ideas so that I can properly allocate tasks and explain the technology.

Week 17 - 6/11/2023

- This week began with some research into Riak to understand the technology before I try to implement it. My knowledge is still shallow on the subject but I believe I understand the relationship between the individual clusters and the nodes inside each. I noted some quirks within the code base itself in how Riak is dealt with, as well with the data store itself. Riak has some problems with speed of retrieval but since the game is based on state based actions in a relatively low tempo environment this probably won't be too much trouble.
- 8th November - Very Ill, making a note here.

Week 18 - 13/11/2023

- This week I am catching up on some of the work that I was unable to complete due to illness last week. Some interesting things that I have learned thus far are the restrictions of Riak within different development environments. For one, it does not seem to be supported for Windows machines which is a problem for my own personal development of the project. I have experimented with VMs however the actual download links through rpm and yum installs all reach dead links. I will need to speak with Hans about this pretty early into Thursday's meeting.
- On a more productive note the faction system for entering and kicking out members has been implemented, and learned some neat things about protocol buffers from the experience. There is still quite a bit of work to be done on this side but my productivity has been greatly increased as I have been experimenting with my workflow and found a pretty successful method for keeping tempo.
- The current situation stands that I am still behind on the faction system but most of the backend support has been implemented such that changes should not require much time to finish, a split between finishing off the faction system and trying to configure the distributed environment should be the next major milestone to aim for.
- Prediction that the faction system should be finished by Monday at the latest.

Week 19 - 20/11/2023

- This week I focused on finishing off the faction system and working on local network integration.
- Main learning points for this was using the Lidgren networking package to configure the local network, this was managed successfully and as of writing (23/11/2023) a server running on my laptop is able to support multiple clients running on my main desktop computer.
- Took another look at the stature bonus from being a member of faction, reconfigured how it works and have implemented this into the system. Still some features of the faction system that I want to implement.